This repository contains the datasets of Channel State Information (CSI), Body-coordinate Velocity Profile (BVP) and Doppler Frequency Spectrum (DFS) for human gesture recognition using WiFi.

Cite the Paper

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Readers can also check the homepage of project Widar3.0 for more details about WiFi sensing.

Dataset Description

Data Type	File Name Format	Description
CSI	id-a-b-c-d-Rx.dat	'**id**': user's id; '**a**': gesture type, '**b**': torso location, '**c**': face orientation, '**d**': repetition number, '**Rx**': Wi-Fi receiver id.
BVP	id-a-b-c-d-suffix.mat	'**id**': user's id; '**a**': gesture type, '**b**': torso location, '**c**': face orientation, '**d**': repetition number. Each file is a **20*20*T** matrix, where the first dimension represents the velocity along x axis ranging between [-2,+2] m/s, the second dimension represents the velocity along y axis ranging between [-2,+2] m/s and the third dimension represents the timestamps with 10Hz sampling rate.
DFS	id-a-b-c-d-suffix.mat	'**id**': user's id; '**a**': gesture type, '**b**': torso location, '**c**': face orientation, '**d**': repetition number. Each file is a **6*121*T** matrix, where the first dimension represents the 6 receivers, the second dimension represents the 121 frequency bins ranging between [-60, 60] Hz and the third dimension represents the timestamps with 1000Hz sampling rate.

Note: gesture type refers to Gesture Description part. Rx location, torso location and face orientation refer to Device Deployment part. Due to space limit, DFS data can be accessed from our FTP.

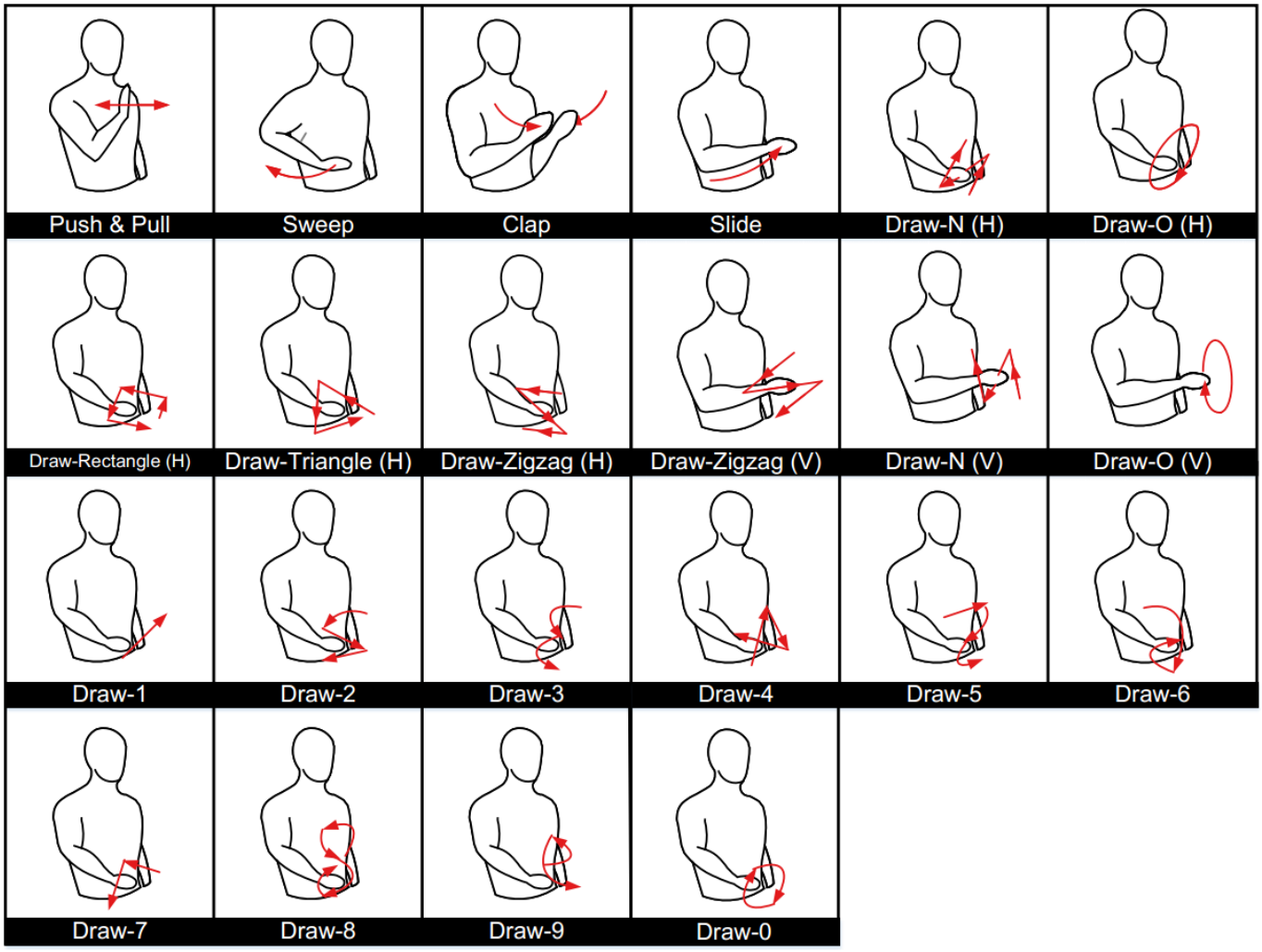
Data Type	Files	Room#	Gestures	Users
CSI	CSI/20181109.zip	1	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-Zigzag(Vertical); 6: Draw-N(Vertical);	User1,2,3
CSI	CSI/20181112.zip	1	1: Draw-1; 2: Draw-2; 3: Draw-3; 4: Draw-4; 5: Draw-5; 6: Draw-6; 7: Draw-7; 8: Draw-8; 9: Draw-9; 0: Draw-0;	User1,2
CSI	CSI/20181115.zip	1	1: Push&Pull; 2: Sweep; 3: Clap; 4: Draw-O(Vertical); 5: Draw-Zigzag(Vertical); 6: Draw-N(Vertical);	User1

Data Type	Files	Room#	Gestures	Users
CSI	CSI/20181116.zip	1	1: Draw-1; 2: Draw-2; 3: Draw-3; 4: Draw-4; 5: Draw-5; 6: Draw-6; 7: Draw-7; 8: Draw-8; 9: Draw-9; 0: Draw-0;	User1
CSI	CSI/20181117.zip	2	1: Push&Pull; 2: Sweep; 3: Clap; 4: Draw-O(Vertical); 5: Draw-Zigzag(Vertical); 6: Draw-N(Vertical);	User4
CSI	CSI/20181118.zip	2	1: Push&Pull; 2: Sweep; 3: Clap; 4: Draw-O(Vertical); 5: Draw-Zigzag(Vertical); 6: Draw-N(Vertical);	User2,3
CSI	CSI/20181121.zip	1	1: Slide; 2: Draw-O(Horizontal); 3: Draw-Zigzag(Horizontal); 4: Draw-N(Horizontal); 5: Draw-Triangle(Horizontal); 6: Draw-Rectangle(Horizontal);	User1,2,3
CSI	CSI/20181127.zip	2	1: Slide; 2: Draw-O(Horizontal); 3: Draw-Zigzag(Horizontal); 4: Draw-N(Horizontal); 5: Draw-Triangle(Horizontal); 6: Draw-Rectangle(Horizontal);	User2,5
CSI	CSI/20181128.zip	2	1: Push&Pull; 2: Sweep; 3: Clap; 4: Draw-O(Horizontal); 5: Draw-Zigzag(Horizontal); 6: Draw-N(Horizontal);	User6
CSI	CSI/20181130_user5_10_11.zip	1	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal); 7: Draw-N(Horizontal); 8: Draw-Triangle(Horizontal); 9: Draw-Rectangle(Horizontal);	User5,10,11
CSI	CSI/20181130_user12_13_14.zip	1	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal); 7: Draw-N(Horizontal); 8: Draw-Triangle(Horizontal); 9: Draw-Rectangle(Horizontal);	User12,13,14

Data Type	Files	Room#	Gestures	Users
CSI	CSI/20181130_user15_16_17.zip	1	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal); 7: Draw-N(Horizontal); 8: Draw-Triangle(Horizontal); 9: Draw-Rectangle(Horizontal);	User15,16,17
CSI	CSI/20181204.zip	2	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal); 7: Draw-N(Horizontal); 8: Draw-Triangle(Horizontal); 9: Draw-Rectangle(Horizontal);	User1
CSI	CSI/20181205.zip	2	[**User2**]1: Draw-O(Horizontal); 2: Draw-Zigzag(Horizontal); 3: Draw-N(Horizontal); 4: Draw-Triangle(Horizontal); 5: Draw-Rectangle(Horizontal); [**User3**]1: Slide; 2: Draw-O(Horizontal); 3: Draw-Zigzag(Horizontal); 4: Draw-N(Horizontal); 5: Draw-Triangle(Horizontal); 6: Draw-Rectangle(Horizontal);	User2,3
CSI	CSI/20181208.zip	2	[**User2**]:1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; [**User3**]:1: Push&Pull;2: Sweep; 3: Clap;	User2,3
CSI	CSI/20181209.zip	2	[**User2**]:1: Push&Pull; [**User6**]:1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal);	User2,6
CSI	CSI/20181211.zip	3	1: Push&Pull; 2: Sweep; 3: Clap; 4: Slide; 5: Draw-O(Horizontal); 6: Draw-Zigzag(Horizontal);	User3,7,8,9
BVP	BVP.zip	All	The gestures correspond to those in CSI data.	All
DFS	DFS.zip	All	The gestures correspond to those in CSI data.	All

Note: The prefix of zip file represents the data collection date. Due to space limit, DFS data can be accessed from our FTP.

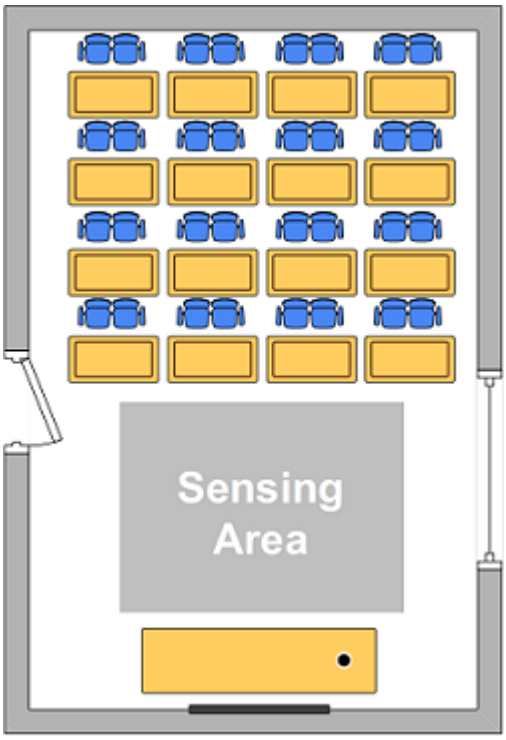
Gesture Description



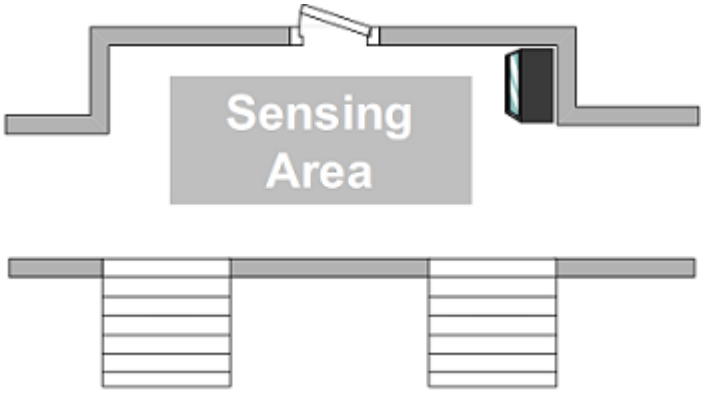
Note: ***(H)*** represents Horizontal and ***(V)*** represents Vertical.

Floor Plan

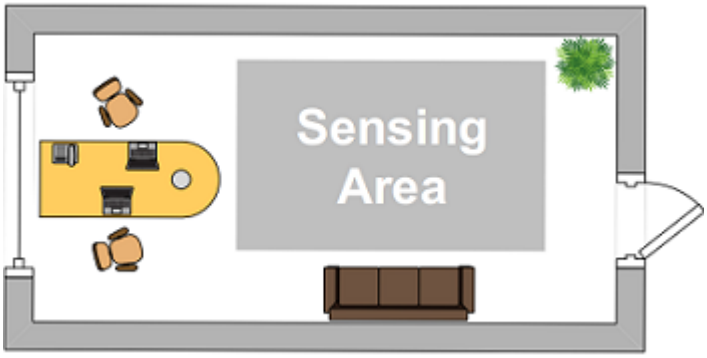
Room#1 - Classroom



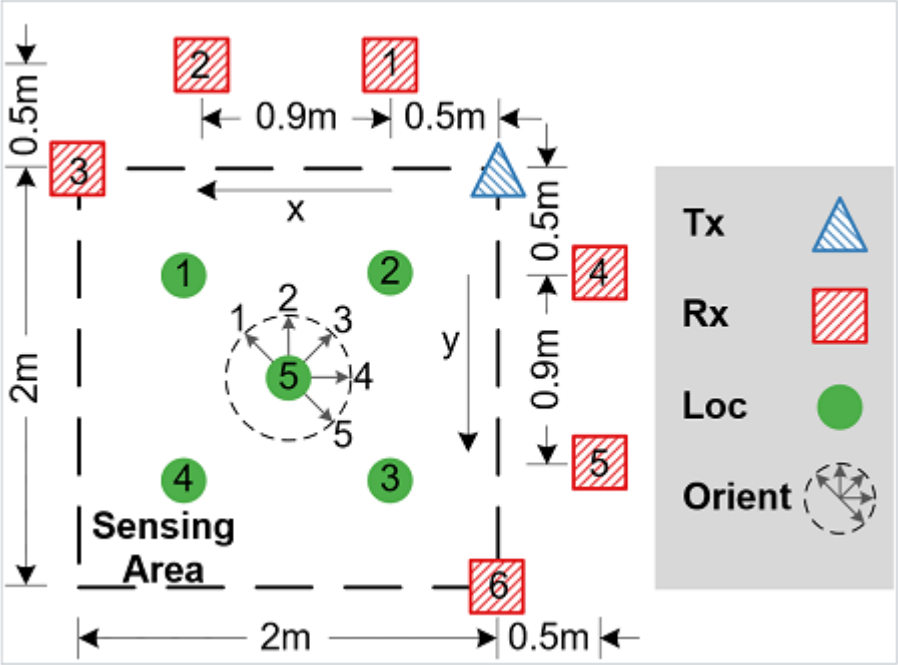
Room#2 - Hall



Room#3 - Office



Device Deployment



Torso Location#	Location (m)	Face Orientation#	Orientation (degree)
1	(1.365, 0.455)	1	-90
2	(0.455, 0.455)	2	-45
3	(0.455, 1.365)	3	0
4	(1.365, 1.365)	4	45
5	(0.91, 0.91)	5	90
6	(2.275, 1.365)		
7	(2.275 2.275)		
8	(1.365 2.275)		

Note: Torso Location #6-8 are not shown in the figure. The reference orientation is denoted as facing the Tx (Orient#3). Tx location is (0, 0).

Code Example

The following shows demo code for BVP extraction and DFS extraction from raw CSI data. Detailed BVP extraction algorithm can be found in our paper.

1. **Extract BVP from CSI**

```
clear;
clc;
close all;

start_index = [1 1 1 1];
total_mo = 1;   % Total motion count
total_pos = 1;  % Total position count
```

```
total_ori = 1;  % Total orientation count
total_ges = 1;  % Total gesture repeatation count
start_index_met = 0;
rx_cnt = 6;     % Receiver count(no less than 3)
rx_acnt = 3;    % Antenna count for each receiver
dpth_pwd = './';
dpth_date = 'Data';
dpth_people = 'userA';
% %%%%
```

```
start_index = [1 1 1 1];
total_mo = 1;   % Total motion count
total_pos = 1;  % Total position count
total_ori = 1;  % Total orientation count
total_ges = 1;  % Total gesture repeatation count
start_index_met = 0;
rx_cnt = 6;     % Receiver count(no less than 3)
rx_acnt = 3;    % Antenna count for each receiver
dpth_pwd = './';
dpth_date = 'Data';
dpth_people = 'userA';
% %%%%
```

Bug Notice

List of empty files

- 20181109/user2/user2-6-4-4-2-r1.dat
- 20181109/user3/user3-1-3-1-8-r5.dat
- 20181118/user2/user2-3-5-3-4-r4.dat
- 20181209/user6/user6-3-1-1-5-r5.dat
- 20181211/user8/user8-1-1-1-1-r5.dat
- 20181211/user8/user8-3-3-3-5-r2.dat
- 20181211/user9/user9-1-1-1-1-r1.dat

Please skip these files when loading them into your applications.